Classification

1: Classification is used when you want to categorize data into different classes or groups. Classification is used when the output is a label instead of a continuous value.

the difference between Classification and Regression Both are supervised learning techniques, but they solve different types of problems depending on the nature of the target variable.

Classification	Regression
Image recognition, sentiment analysis, fraud detection	Predicts a continuous numeric value
Predicts a categorical label	Fitting the best-fit line or curve
Forecasting, price prediction, risk estimation	Drawing a decision boundary between classes

- Real-life example
- Classification Example: predicts categories or labels like disease/no disease, etc.
- Regression Example: Predicting the market value of a used car based on features like manufacturing year and engine capacity.

2: Logistic Regression is a linear classification algorithm that estimates the probability of a data point belonging to a particular class using the sigmoid function.

Advantages

- Easy to implement and interpret.
- Performs well on linearly separable data.
- Less prone to overfitting with regularization.
- Probabilistic output helps in decision-making.

Limitations

- Struggles with non-linear relationships.
- Sensitive to outliers.
- Assumes linear relationship between features and log-odds.

Decision tree is a tree-structured classification algorithm where internal nodes represent feature tests, branches represent decision rules and leaf nodes represent class labels. It mimics human decision-making logic.

Advantages

- Easy to visualize and interpret.
- Handles non-linear relationships.
- Works well with mixed data types.

Limitations

- Prone to overfitting.
- Sensitive to small changes in data.
- Can become complex with deep trees.

Random Forest is an ensemble classification algorithm that builds multiple decision trees using random subsets of data and features and combines their predictions through majority voting.

Advantages

- High accuracy and robustness.
- Handles large datasets well.
- Reduces overfitting compared to single trees.

Limitations

- Less interpretable than decision trees.
- Computationally intensive.
- Large models consume more memory.

3: Classification problems aim to predict discrete categories. To evaluate the performance of classification models, we use the following metrics:

- Accuracy: is a fundamental metric used for evaluating the performance of a classification model.
- Precision It measures how many of the positive predictions made by the model are actually correct.
- Recall or Sensitivity measures how many of the actual positive cases were correctly identified by the model.
- F1-Score is the harmonic mean of precision and recall. It is useful when we need a balance between precision and recall as it combines both into a single number.

- Confusion Matrix creates a N X N matrix, where N is the number of classes or categories that are to be predicted.

Metrics	**Measure**	**Sensitivity to class imbalance**
Precision	Reliability of positive prediction	Robust
F1-Score	Balance between precision and recall	Very Robust
Confusion	A 2 x2	Extremely Robust
Accuracy	Overall correctness of the model	Highly sensitive
Recall	Ability to capture all true positive	Robust

4: Class imbalance occurs when the target variable contains classes that are not equally represented.

Why Accuracy Becomes Misleading

Accuracy assumes that all errors are equally important. In imbalanced datasets, this assumption fails.

prioritize Recall: when the goal is to approve as many qualified loan applicants as possible even if a few risk applicants are also approved.

prioritize precision: when the goal is to ensure that only truly qualified are approved, reducing the risk of giving loans to people who may not repay.

5: Real-World Case Study: Loan Approval Prediction

Goal:

The goal was to predict whether a loan application should be approved or rejected based on customer information.

Data Used:

The dataset included applicant income, loan amount, credit history, employment status, education level, and marital status.

Type of Classification Applied:

Random Forest Classifier was used to classify applicants into two categories: Approved or Rejected.

Key Results/Insights:

The Random Forest model achieved high accuracy and was able to identify eligible applicants more effectively than a simple model. It also reduced the number of incorrect loan approvals, helping banks make better lending decisions.